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RESEARCH ARTICLE

RFMVDA: An Enhanced Deep Learning Approach for Customer Behavior Classification in E-Commerce Environments

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ABSTRACT Customer Relationship Management (CRM) systems, widely used in enterprises, have evolved into Software-as-a-Service (SaaS) platforms. With the advent of Customer Data Platforms (CDP), these systems continuously store customer behavior data for purposes such as creating single customer profiles, analyzing, tracking, and managing customer interactions from various perspectives. With the global expansion of the e-commerce market, research on customer analysis and classification optimized for the e-commerce environment has been actively conducted. The RFM (Recency, Frequency, Monetary) model is a straightforward method for classifying customers and is applied across various industries. However, in the e-commerce environment, where customers can access services at any time, there are limitations in collecting, storing, and reflecting customer behavior data for classification. To resolve these limitations, this paper proposes the RFMVDA (Recency, Frequency, Monetary, Visits, Durations, Actions) model. This model is designed to capture customer data, sessions, and behavior units suitable for the e-commerce environment. By utilizing the RFMVDA model for customer behavior-based segmentation and classification, we constructed a Deep Neural Network (DNN) to predict customer behavior-based classifications. As a result, the proposed model demonstrated a segmentation prediction accuracy of 92.98% for customers in the e-commerce environment.

INDEX TERMS Customer segmentation, customer classification, machine learning, deep neural network (DNN), customer data platform (CDP), customer relationship management (CRM).

I. INTRODUCTION

As computing power has increased, big data and Artificial Intelligence (AI) have become integral to our daily lives. With advancements in big data processing technologies, various industries have increasingly adopted methods for data storage and management. According to global market research firm IDC, the volume of data is expected to grow from 33ZB in 2018 to 172ZB by 2025, with an annual growth rate of 61% [1]. Many companies have seen an accelerated growth in digital business and data complexity,

leading to a high demand for data ETL (Extract, Transform, Load) processes across various industries. This has resulted in the widespread use of big data platforms for storing and managing domain-specific data [2]. Companies use Customer Relationship Management (CRM) software to handle complex and diverse data. CRM systems perform ETL on customer data and maintain this information over long periods for business management purposes. CRM is crucial for managing customer interactions, enhancing sales, creating value, and personalizing products and services [3], [4], [5], [6], [7]. Recently, companies have begun using Customer Data Platforms (CDP) to continuously store information about customers who access their services and

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websites. These platforms allow for the creation, tracking, and management of single customer profiles, analyzing customer behavior from various perspectives [8]. Various studies on customer segmentation have been conducted using the collected customer data. A representative method for customer segmentation is the RFM (Recency, Frequency, Monetary) model [9], [10], [11]. Companies use the RFM model to classify existing customers, thereby enabling targeted marketing and promotions to encourage repeat visits or purchases. The RFM model categorizes customers based on recency, frequency, and monetary value of purchases [12]. However, the traditional RFM model has significant limitations, especially in the context of modern e-commerce environments.

Firstly, the traditional RFM model is only capable of classifying customers who have made a purchase. It does not account for customers who visit a store or platform without making a purchase, thereby overlooking a significant segment of customer interactions. Secondly, the RFM model is traditionally applied in offline retail environments where customer interactions are bound by store hours, and the data collected is limited to those specific timeframes. This approach is inadequate for e-commerce, where customer interactions are continuous and can occur at any time. Thirdly, the RFM model is not optimized for online e-commerce platforms. In e-commerce, customers engage in various activities beyond just purchasing, such as browsing, adding items to carts, and spending time on specific product pages. These activities form a crucial part of the customer journey, which the RFM model fails to capture.

To address these challenges, this paper proposes the RFMVDA (Recency, Frequency, Monetary, Visits, Durations, Actions) model, which extends the traditional RFM model by incorporating additional dimensions that are crucial for capturing customer behavior in e-commerce environments. The RFMVDA model introduces three new attributes—Visits, Durations, and Actions—allowing for a more comprehensive analysis of customer interactions. By integrating these additional behavioral dimensions, the RFMVDA model not only enhances the granularity of customer segmentation but also makes it possible to classify customers based on their entire journey on an e-commerce site, including those who do not complete a purchase. This approach ensures that businesses can gain a deeper understanding of both purchasing and non-purchasing behaviors, enabling more effective and targeted marketing strategies.

Furthermore, we have chosen to implement a Deep Neural Network (DNN) model to process the high complexity and volume of data associated with e-commerce platforms. DNNs are particularly well-suited for this task due to their ability to model complex patterns and interactions within large datasets, making them ideal for customer classification tasks that involve multiple behavioral dimensions. The contributions of this research lie in the development of the RFMVDA model and the application of DNN for enhanced

customer classification, both of which significantly improve upon traditional methods.

The structure of this paper is as follows: Section II introduces concepts for customer segmentation and related studies on customer behavior and purchase prediction in the e-commerce environment. Section III proposes the RFMVDA model for customer behavior data in e-commerce, including segmentation for non-purchasing customers. Section IV evaluates and analyzes the learning accuracy and prediction accuracy for customer behavior classification using the proposed DNN model. Finally, Section V provides conclusions and future research directions.

II. RELATED WORK

The RFM (Recency, Frequency, Monetary) model is one of the most widely used methods for customer segmentation and analysis in marketing. It simplifies the classification of customers based on three criteria, making it an effective tool for marketing applications. The RFM model has been employed in industries for over 30 years as a direct marketing approach and remains widely used due to its ease of implementation. Companies use the RFM model to classify customers into categories such as loyal customers, potential customers, and lost customers, thereby gaining insights into consumer loyalty and relationships with the company's products [11]. The RFM model ranks customers based on the recency, frequency, and monetary value of their purchases, converting this information into a two-dimensional data format. Subsequent clustering algorithms like K-means and fuzzy inference are then applied to segment customers [13], [14], [15]. The LRFM model extends the RFM model by adding a Length factor, representing the time span between the first and last purchases [10]. This model measures the purchase cycle of customers, enabling further segmentation into core customers, potential customers, lost customers, and new customers based on the Length attribute.

In the e-commerce environment, customer segmentation and purchase prediction research often utilize Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks to learn customer behavior and predict purchase likelihood [16], [17]. Studies combining LSTM with Random Forest (RF) algorithms analyze customers' purchasing activities and access patterns over time to predict consumer behavior. These models input customer behavior and purchase history into an LSTM-RF model, producing purchase probability outputs through a Fully Connected Layer [18]. Although LSTM-RF models have shown high performance in predicting customer purchases in e-commerce, they mainly focus on customers who have made purchases, thus limiting their ability to classify non-purchasing user behaviors.

The Self-Organizing Neural Network (SONN) employs deep learning for customer segmentation in digital marketing within the e-commerce environment [19]. This unsupervised learning model clusters customer data using the RFM model, SOM (Self-Organizing Map), and DNN (Deep Neural

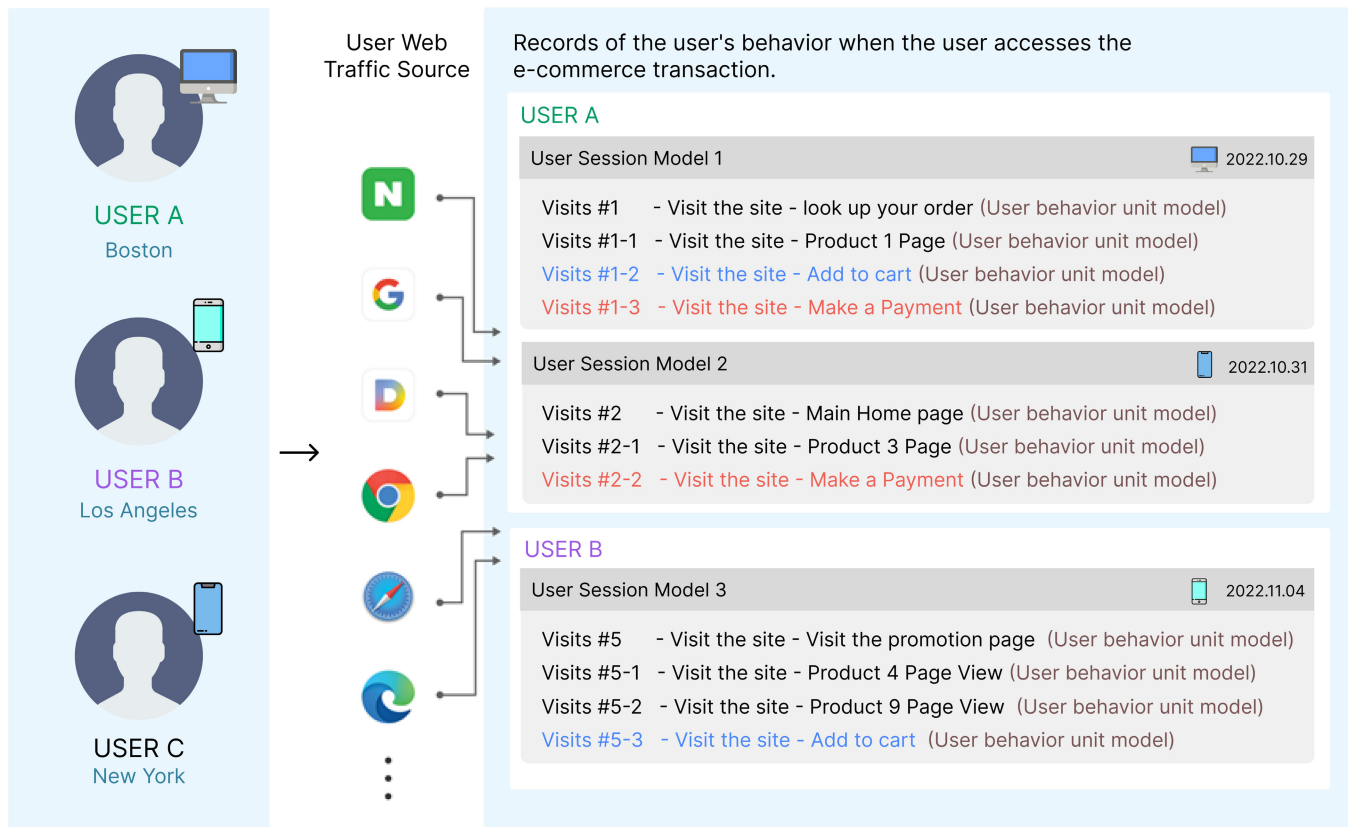


FIGURE 1. Flow diagram of customer information, session, and behavior unit data modeling.

Network). The SONN model classified 4,372 customers into five segmented groups with an accuracy of 98%. SONN's ability to generate two-dimensional data clusters allows for easy identification of relationships and patterns in the data. However, the results can vary depending on the similarity and distance functions used, increasing model complexity. Moreover, SONN primarily considers purchase data for customer classification, excluding factors like access environment, behavior data, dwell time, and purchased products, which are essential for behavior-based segmentation in the e-commerce environment.

Recent studies have further explored the application of machine learning techniques for client segmentation in e-commerce settings. For instance, Banerjee and colleagues [20] examined AI-driven approaches for customer profiling, segmentation, and sales prediction in direct marketing. Their research highlights how integrating AI techniques can improve segmentation accuracy and predict sales trends, thereby significantly enhancing personalized marketing strategies. Similarly, Sriprasad and co-authors [21] investigated the use of various machine learning algorithms for client segmentation and customization within e-commerce environments. Their approach included data preprocessing using Python's TensorFlow and Pandas libraries, followed by the application of clustering models such as K-means, DBSCAN, and Agglomerative Clustering.

The study achieved an accuracy of approximately 93% in identifying target customer segments, underscoring the potential of these techniques to enhance customer segmentation strategies and enable more effective personalized marketing. Additionally, Vijai et al. [22] applied a range of machine learning algorithms, including clustering and predictive modeling techniques, for customer segmentation in e-commerce management. Their study emphasizes the critical role of personalized marketing strategies tailored to distinct customer segments.

These advancements highlight the ongoing evolution of customer segmentation techniques, particularly in adapting traditional models to the complexities of the e-commerce environment.

III. ECBC (EFFICIENT CUSTOMER BEHAVIOR CLASSIFICATION)

A. CUSTOMER DATA MODELING

This paper collects various customer behavior data (such as access information, webpage visits, dwell time, and device usage) in the e-commerce environment and uses it to segment customers through machine learning techniques. Customer data is modeled into three main components: customer information, sessions, and behaviors. Based on this model, the data is transformed into attributes necessary for customer segmentation in the proposed RFMVDA (Recency,

TABLE 1. Definition of RFMVDA attributes.

Variable	Definition
Recency	Time since the last purchase
Frequency	Number of purchases in a given period
Monetary	Total amount spent by the customer
Visits	Total number of visits in a given period
Durations	Total dwell time on the site (in seconds)
Actions	Total actions (page views, cart additions, etc.)

Frequency, Monetary, Visits, Durations, Actions) model. This transformed data is used to classify and segment customers using machine learning models and Deep Neural Networks (DNNs) designed for the e-commerce environment. When a customer accesses a specific e-commerce platform or website, information such as the entry channel (various search engines, advertisements, etc.), device, region, country, and time is transmitted to and stored on the server. This information is contained in web access information headers, which are standardized by web protocols. All information about customer behaviors and transaction histories from the moment they enter the e-commerce environment until they leave is stored on the server. The stored customer information is categorized into three data models: customer information model, session model, and behavior unit model, as shown in Figure 1. The customer information model records basic identification data of the customer accessing the e-commerce platform, including device, region, browser, and operating system data. This model is updated each time the customer accesses the website, storing changes in the user's access environment in real-time. The session model comprehensively records customer behavior during their stay on the website. It includes information about each session generated when the customer visits the site, such as start time, end time, and page sequence, to store data on the customer's e-commerce site activities. This data is used to analyze detailed information such as which products the visitor showed interest in and which pages they spent more time on. The behavior unit model stores specific behavior information during the customer's journey on the site. It segments and stores actions such as each page view, number of clicks, cart additions, and purchase activities, which are essential for subsequent customer segmentation. The unique ID assigned to the customer links the session model and individual behavior units, storing interactions related to customer preferences, interests, and purchasing tendencies. This model integrates with the session model, connecting all behavior units occurring in each session into a single session.

B. RFMVDA SEGMENTATION MODEL (PURCHASING CUSTOMER SEGMENTATION)

Customers generally access various pages when they visit an e-commerce platform. In the case of a shopping mall's e-commerce site, customers tend to browse detailed information about products and contemplate making purchases.

TABLE 2. Definition of VDAR attributes.

Variable	Definition
Visits	Total number of visits in a given period
Durations	Total dwell time on the site (in seconds)
Actions	Total actions (page views, cart additions, etc.)
Referral Keyword	Indicator of whether a keyword was used in the referral (0 or 1)

Consequently, they spend time on the product detail pages, increasing their dwell time on the e-commerce website. To classify customers in the e-commerce environment, the RFMVDA (Recency, Frequency, Monetary, Visits, Durations, Actions) model adds three attributes—visits, durations, and actions—to the traditional RFM model, resulting in a total of six attributes, as shown in Table 1. Customers tend to spend time on product detail pages while comparing products of interest. When customers view or purchase products on an e-commerce platform, the number of visits to the shopping mall's e-commerce site increases. Additionally, if customers are satisfied with the purchased products, they will continue to access the e-commerce site to browse or purchase other products. Therefore, the number of visits is a crucial attribute for classifying customers in the e-commerce environment. If a customer decides that the content of the shopping mall's e-commerce website is unnecessary, the time they spend viewing the detail pages will decrease. For instance, if a customer accesses the e-commerce site via a search engine using the keyword "60-inch TV," it is highly likely that the customer is interested in purchasing a product related to the keyword. The customer will browse the products of interest or comparable products on the e-commerce site and spend time on the detail pages. Hence, the total dwell time is an essential indicator of the customer's decision to purchase a product. Lastly, the action attribute reflects customer behavior such as repetitive page views and adding products to the cart. These actions indicate satisfaction with the purchased products, leading to continuous access to the site to browse or purchase other products. Therefore, the total number of visits, total dwell time, and total purchase actions are critical indicators for customer classification in the e-commerce environment.

C. VDAR SEGMENTATION MODEL (NON-PURCHASING CUSTOMER SEGMENTATION)

Traditional RFM models are limited in that they primarily focus on segmenting customers based on purchasing behavior, leaving non-purchasing customers unanalyzed. However, in e-commerce environments, customers can freely access the site at any time to browse products without necessarily making a purchase. Unlike physical stores, where it is difficult to track non-purchasing customers' activities, e-commerce platforms can capture and model a wide range of behavioral data, regardless of whether the customer makes a purchase.

To address the gap in customer segmentation for non-purchasing customers, this study proposes the VDAR (Visits,

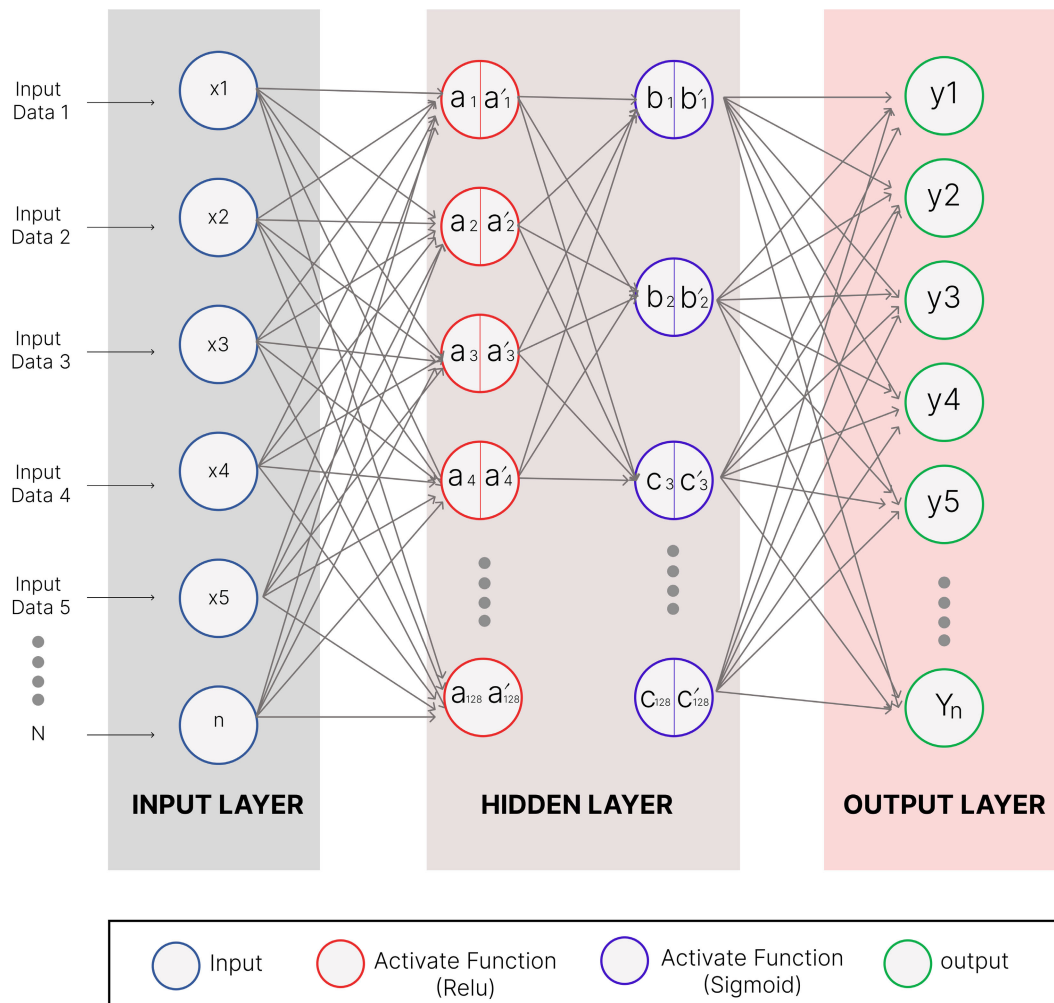


FIGURE 2. DNN Configuration for customer classification.

Durations, Actions, Referral Keyword) model. The VDAR model classifies customers based on four key attributes: *Visits*, *Durations*, *Actions*, and *Referral Keyword*, as summarized in Table 2. These attributes enable the model to assess customer engagement by analyzing their site visit frequency, session duration, behavioral patterns, and their referral sources before landing on the e-commerce site.

For instance, consider a customer who accessed the e-commerce site after searching for “60-inch TV” on a search engine. This customer is likely to have a high interest in purchasing a television or gathering information about similar products. The customer may browse various product models, read reviews, and compare prices, leading to an increase in their total session duration and page views. If the product details and pricing meet their expectations, they may revisit the site more frequently, and their visit count and page view count would naturally increase. Additionally, the referral keyword attribute provides insight into the customer’s intent based on the search terms they used before landing on the e-commerce site.

The VDAR model’s classification of non-purchasing customers, based on their behavioral attributes, is crucial

for companies aiming to derive actionable insights from customer activities beyond purchase behavior. This model helps marketers and decision-makers identify customers who are in the consideration phase, allowing them to tailor targeted marketing strategies that could lead to future purchases. For example, if a promotion generates high site traffic but low engagement in terms of page views and session durations, it could indicate that the products or content may not be resonating with the audience, providing valuable feedback for future optimizations. Therefore, the VDAR model enables the segmentation of non-purchasing customers based on their behavioral data and serves as a key tool for improving overall marketing efficiency and driving data-driven decision-making in e-commerce businesses.

Furthermore, by applying the VDAR model, businesses can better understand and respond to the entire customer journey on e-commerce platforms. This model not only captures purchase-related behavior but also provides insights into the broader context of user interactions, including those that do not result in immediate transactions. This comprehensive approach allows companies to tailor their strategies to engage potential customers more effectively

TABLE 3. Definition of input layer parameters.

No	Data Model Type	Variable Name	Descriptions
1	Customer Data	Total_spent_time	Total time spent
2	Customer Data	Total_unique_action	Number of unique actions
3	Customer Data	Total_more_than_once	Number of pages viewed more than once
4	Customer Data	Total_revenue	Total purchase amount
5	Customer Data	Total_items	Total number of items purchased
6	Customer Data	Total_ecommerce	Total number of orders
7	Customer Data	First_visit	Date and time of first visit
8	Customer Data	First_referrer_name	First referral source
9	Customer Data	First_referrer_keyword	First referral keyword
10	Customer Data	Last_visit	Date and time of last visit
11	Customer Data	Last_referrer_name	Last referral source
12	Customer Data	Last_referrer_keyword	Last referral keyword
13	Customer Data	Total_desktop_visit	Number of visits via desktop
14	Customer Data	Total_mobile_visit	Number of visits via mobile
15	Session Data	Location_country	Country of the session
16	Session Data	Location_city	City of the session
17	Session Data	Config_browser_name	Browser used in the session
18	Session Data	Device_type	Type of device used in the session
19	Session Data	Visit_first_action_time	Time of first action in the session
20	Session Data	Visitor_seconds_since_first	Time since first visit in seconds
21	Session Data	Visitor_seconds_since_last	Time since last visit in seconds
22	Session Data	Visitor_seconds_since_order	Time since last order in seconds
23	Session Data	Visit_total_time	Total time spent in the session
24	Session Data	Visit_total_actions	Total number of actions in the session
25	Session Data	Referrer_name	Referral source for the session
26	Session Data	Referrer_keyword	Referral keyword for the session
27	Session Data	Referrer_url	Referral URL for the session
28	Session Data	Total_items	Total number of items purchased during the session
29	Session Data	Total_revenue	Total revenue from the session

throughout their online journey, from initial site visits to eventual purchases.

D. DNN CONFIGURATION FOR CUSTOMER CLASSIFICATION

The customer data accessed in the e-commerce environment is processed and normalized according to the RFMVDA model, and a neural network is constructed to classify customers. The configuration of the DNN is shown in Figure 2. The input layer consists of N data points from the customer's session model. The hidden layer is composed of two layers, each connected by neurons. The input data is passed to the hidden layers, with the first hidden layer consisting of 128 neurons activated by the ReLU activation function. The second hidden layer also consists of 128 neurons, connected by the sigmoid activation function. Finally, the output layer receives the final result.

We selected a Deep Neural Network (DNN) for this task due to its ability to model complex relationships within large datasets, which are common in e-commerce environments. DNNs excel at capturing intricate patterns and dependencies in data, making them particularly suitable for tasks such as

customer classification, where numerous behavioral factors must be considered simultaneously. Furthermore, DNNs can efficiently handle the high dimensionality and volume of data associated with the RFMVDA model, ensuring robust and accurate predictions.

The deep learning model is trained to predict values as close as possible to the actual values by calculating the error loss when the predicted value differs from the actual value during forward propagation. Various loss functions exist, with binary cross-entropy used when the output values are 0 or 1. However, this is not suitable for multi-class classification where there are more than two classes.

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C t_{ij} \log(y_{ij}) \quad (1)$$

Categorical cross-entropy is used for multi-class classification problems, as indicated by Equation (1), where C represents the number of classes. Therefore, this paper employs categorical cross-entropy as the loss function since the classification model is trained on data consisting of 15 different categories for purchasing customers and 9 different categories for non-purchasing customers. Finally,

TABLE 4. Composition of the performance evaluation dataset.

Indicators	Description and Data Values
E-Commerce Site Categories	Wholesale and retail sales of daily necessities
Data Collection Period	January 1, 2022 – March 31, 2022
Total Registered Members	9,416 individuals
Customers Who Accessed the Site Within the Period	5,041 individuals
Customers Who Purchased Within the Period	299 individuals
Customers who visited the site and did not purchase within the period	4,742 individuals
Total Number of Visits by All Customers (Sessions by Customers)	37,698 visits
Total Number of Customer Activities (Total Customer Pageviews, Last-Minute Purchase Activity, etc.)	405,280 activities
Quantity of Products Purchased	10,892
E-commerce Sales Amount	₩92,177,675

TABLE 5. Customer segmentation results using LRFM model.

Customer Classification Name	Number of Customers (Total 299)	Percentage (%)
VIP Customers	91	30%
Potential Loyal Customers	39	13%
Regular Customers	29	10%
Lost Customers	140	47%
Total	299	100

a DNN is constructed, as shown in Figure 2, to standardize and train the RFMVDA data model.

E. INPUT LAYER PARAMETERS FOR DNN

The parameters of the data model used in the input layer of the DNN are configured as shown in Table 3. These parameters include 14 customer data points and 15 session data points, totaling 29. These parameters serve as essential input data for customer classification and behavior prediction through the DNN. Each time a customer accesses the e-commerce website, a unique session identifier is created. During the session, if the customer browses various product pages and makes purchases, data such as the session's dwell time, total number of actions, total purchase quantity, and total purchase amount is stored. When the customer exits the e-commerce website, the session information in the customer data model is updated with the final actions taken by the customer. This process of storing and compiling customer actions from the moment they access the website until they exit provides the 29 parameters, which are then used to train the DNN.

IV. PERFORMANCE EVALUATION

A. DATASET

The dataset used in this study is based on real customer data collected from an e-commerce website in South Korea. This dataset spans a 3-month period from January 1, 2022, to March 31, 2022, and includes information on a total of 9,416 registered customers. Among these, 5,041 customers visited the website, and 299 completed a purchase. The dataset includes various data such as customer registration details, website visit logs, session information from each visit, and customer activity records. In total, there are

37,698 customer sessions and 405,280 customer activity data points. The dataset primarily contains variables for analyzing customer behavior, including visit time, page views, cart additions, and product purchases, as summarized in Table 4. These behaviors were the focus of analysis and were used to evaluate the proposed RFMVDA model, as well as to compare it against the LRFM model.

The raw data collected from the e-commerce environment contains unstructured data that requires refinement and preprocessing for analysis. Initially, irrelevant features and missing data were removed, followed by normalization of the data into a suitable format for analysis. Given the diverse distribution of customer behavior data, each variable's values were scaled between 0 and 1 to facilitate input into the model.

Throughout this process, the RFMVDA model was constructed to analyze each customer's behavior pattern in detail. This model includes six key variables: Recency, Frequency, Monetary, Visits, Durations, and Actions. For the purposes of model training and testing, the dataset was split into two sets: 70% was used for training, while the remaining 30% was reserved for testing.

B. CUSTOMER SEGMENTATION BASED ON LRFM MODEL

The LRFM model was applied to analyze purchasing customer behavior patterns and to categorize customers into four segments. This model assigns scores to each customer based on four factors: Length (the time between the first and most recent purchase), Recency (the time since the most recent purchase), Frequency (the number of purchases), and Monetary (the total amount spent). Using these four elements, the customers were classified into specific segments.

TABLE 6. Final customer segmentation applied using the RFMVDA model.

Customer Classification Name	Number of Customers (Total 299)	Percentage (%)
VIP Customers	73	24.4
Loyal Customers	57	19.1
Promising Loyal Customers	9	3.0
Potential Loyal Customers	30	10.0
Regular Customers	0	0.0
Promising Customers	1	0.3
Customers Needing Attention	0	0.0
About to Sleep Customers	47	15.7
At Risk Customers	24	8.0
About to Leave Customers	13	4.3
Customers to Retain	0	0.0
Reconsideration Customers	2	0.7
Hibernating Customers	14	4.7
Lost Customers	29	9.7
Total	299	100

Table 5 shows the customer segmentation after applying the LRFM model. A total of 299 customers were classified into four primary segments: VIP Customers, Potential Loyal Customers, Regular Customers, and Lost Customers. Each segment was determined by factors such as purchase frequency, the recency of purchases, total monetary value, and the length of time from the first visit to the purchase.

The VIP customers account for 30% of the total 299 customers. These customers recently made frequent purchases with high monetary value and are considered very important. On the other hand, the Lost Customers group comprises 47% of the total customers. This group requires further customer relationship management strategies to either retain or reactivate them. The VIP Customers group maintains recent and frequent purchases with high spending, placing them at the top of the LRFM model's scoring. These customers represent a loyal base for the business, and offering special promotions or benefits can maximize retention and engagement. Potential Loyal Customers, although having lower purchase frequency and spending than VIP customers, have still made recent purchases and hold the potential to be converted into VIP Customers. They are a key target for additional marketing strategies aimed at customer conversion. Regular Customers are those with lower purchase frequency but consistent business engagement. Custom marketing strategies can be applied to reinforce their buying patterns and encourage more frequent purchases. Lost Customers are those who have not made recent purchases and exhibit low frequency and spending patterns. This segment requires targeted remarketing campaigns or incentives to encourage repurchases and re-engagement with the business.

Although the LRFM model successfully segmented customers based on their purchasing behavior, it has certain limitations, particularly in the e-commerce environment. This model fails to account for the activities of customers who have not made any purchases. For instance, even if customers visit the site and engage in various activities, if they do not

make a purchase, the LRFM model does not capture this behavior. Thus, a more comprehensive model that includes non-purchasing customers is needed.

Additionally, the model does not fully reflect all customer access activities in the e-commerce environment. Since customers can engage in a variety of actions at any time, it is critical to evaluate the potential likelihood of these actions leading to purchases. To overcome these limitations, it is necessary to develop an expanded model that can analyze a broader range of customer access patterns and behavioral data.

C. CUSTOMER SEGMENTATION BASED ON THE RFMVDA MODEL

The RFMVDA model extends the traditional LRFM framework to enhance the granularity of customer behavior analysis and segmentation in the e-commerce environment. In addition to the standard Recency (R), Frequency (F), and Monetary (M) dimensions, the RFMVDA model introduces three additional variables—Visits (V), Durations (D), and Actions (A). These supplementary dimensions enable a more comprehensive and detailed analysis of customer behavior across a broader spectrum of interactions within the e-commerce platform.

Unlike traditional models that focus primarily on purchase history, the RFMVDA model provides a more holistic view by incorporating various online activities. This allows for a richer segmentation of customers based on their engagement with the platform, taking into account not only their purchasing behavior but also metrics such as visit frequency, time spent on pages, and specific actions performed on the site. To ensure that each behavioral dimension contributes equally to the analysis, the data for each variable is normalized using Min-Max scaling to map the values between 0 and 5. This standardization enables the model to treat all behavioral attributes with

TABLE 7. Detailed customer segmentation transition from LRFM to RFMVDA model.

Customer Count (Total 299)	Customer Segmentation Name (LRFM Model)	Modified Customer Count	Customer Segmentation Name (RFMVDA Model)
91	VIP Customers	60	VIP Customers
		20	Loyal Customers
		2	Noteworthy Potential Loyal Customers
		4	Potential Loyal Customers
		5	At Risk Customers
39	Potential Loyal Customers	13	VIP Customers
		17	Loyal Customers
		4	Noteworthy Potential Loyal Customers
		5	Potential Loyal Customers
29	Regular Customers	7	Loyal Customers
		5	Potential Loyal Customers
		11	At Risk Customers
		2	Reconsideration Customers
		4	About to Leave Customers
140	Lost Customers	13	Loyal Customers
		3	Noteworthy Potential Loyal Customers
		16	Potential Loyal Customers
		1	Promising Customers
		8	At Risk Customers
		47	About to Sleep Customers
		9	About to Leave Customers
		29	Lost Customers
		14	Hibernating Customers

equal importance, thereby providing a balanced evaluation of customer engagement.

Through the application of the RFMVDA model, customers were classified into 14 distinct segments, as shown in Table 6. This segmentation approach offers a significantly more nuanced classification compared to the traditional LRFM model, as it accounts for a wider range of customer behaviors. The RFMVDA model not only considers purchasing habits but also integrates data on overall site interactions, providing a more detailed and accurate reflection of customer behavior patterns.

This refined segmentation captures the complexity of customer interactions within the e-commerce environment. The RFMVDA model offers a more precise categorization by considering both purchase history and diverse behavioral metrics, thereby allowing businesses to better tailor their marketing strategies and customer relationship management efforts.

D. COMPARATIVE ANALYSIS OF LRFM AND RFMVDA MODELS

The segmentation results of 299 customers who made purchases in the e-commerce environment were compared using both the LRFM and RFMVDA models. The LRFM model classifies customers into four simple segments: VIP Customers, Potential Loyal Customers, Regular Customers, and Lost Customers. In contrast, the RFMVDA model is designed to account for various customer behaviors in the e-commerce environment by incorporating not only

purchasing behavior but also variables such as Visits, Durations, and Actions, leading to the segmentation of customers into 14 detailed categories.

1) COMPARISON OF CUSTOMER SEGMENTATION RESULTS
The LRFM model, based on relatively simple metrics, is useful for identifying key customer groups but has limitations in fully reflecting the complex user behavior patterns found in dynamic online environments like e-commerce. For example, in the LRFM model, 140 customers were classified as Lost Customers due to a significant decrease in their purchase frequency or length attributes, even though they may have completed purchases during other periods. In the LRFM model, ‘lost customers’ specifically refer to those whose purchasing activity has declined significantly within the defined analysis period, rather than customers who have never made a purchase at any point.

In contrast, the RFMVDA model considers additional variables such as visit frequency, session duration, and action counts to further segment these customers into categories such as About to Sleep Customers, At Risk Customers, and About to Leave Customers. This approach allows for a more detailed classification of customers, identifying early signs of churn that the LRFM model may overlook.

2) MODEL COMPARISON ANALYSIS
The LRFM model is useful for providing a simple customer classification based on purchasing data. However, it falls short of reflecting comprehensive online behavior data.

TABLE 8. Detailed customer segmentation based on VDAR model.

Customer Segmentation Name	Number of Customers (Total 4,742)	Percentage (%)
Customers likely to purchase soon	124	3%
Highly engaged customers	775	16%
Customers considering a purchase	139	3%
Content consumers	673	14%
Browsing various products	1,097	23%
Conservative buyers	32	1%
At risk of abandoning	929	20%
Disengaged customers	638	13%
Misguided visitors	335	7%
Total	4,742	100%

In contrast, the RFMVDA model provides the ability to segment customers by incorporating multidimensional behavioral data from the e-commerce environment. By including non-purchase activities, such as visit frequency, duration, and action counts, the RFMVDA model offers a more complete evaluation of customer engagement, allowing for more accurate predictions of potential customer churn.

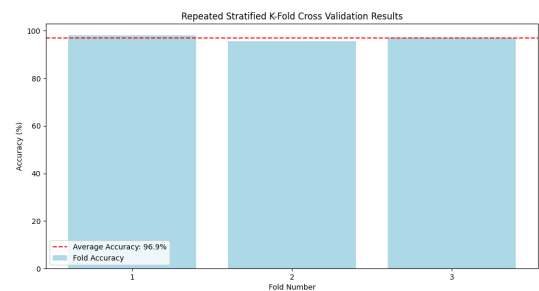
For example, as shown in Table 7, 16 of the 140 customers categorized as Lost Customers in the LRFM model were reclassified as Potential Loyal Customers in the RFMVDA model due to the inclusion of visit frequency and site activity data. This highlights the ability of the RFMVDA model to create more precise and effective customer segmentation, allowing businesses to implement more detailed CRM strategies. By identifying customers who show early signs of churn, targeted marketing and reactivation campaigns can be launched to improve customer retention.

In conclusion, the RFMVDA model is better suited for the e-commerce environment as it incorporates a broader range of behavioral data, making it a powerful tool for preventing customer churn and developing effective marketing strategies that target various customer segments based on their detailed behavior profiles.

E. BEHAVIORAL SEGMENTATION OF NON-PURCHASING CUSTOMERS USING THE VDAR MODEL

As shown in Table 8, the VDAR model was applied to segment non-purchasing customers who visited the e-commerce platform during the study period from January to March 2022. Of the 5,041 total visitors, 4,742 customers did not complete a purchase. The VDAR model enabled a detailed segmentation of these non-purchasing customers, taking into account their behavioral data such as total visits (Visits), total dwell time (Durations), total actions (Actions), and the presence of referral keywords (Referrer).

The analysis revealed that approximately 36% (1,711 customers) of the non-purchasing customers fell into the top four segments most likely to convert to purchasing customers. These segments include customers who are likely to purchase soon and customers showing high engagement with the content on the site, such as viewing multiple

**FIGURE 3.** Repeated stratified K-Fold cross validation results.

products or spending significant time on product detail pages. Additionally, 1,097 customers (22%) demonstrated a keen interest in various products, indicating strong potential for future purchases. However, 1,902 customers (40%) were classified in the lower engagement segments, indicating a higher risk of disengagement or abandonment.

Marketers and decision-makers can utilize this segmentation to develop personalized promotional strategies aimed at converting the top segments into first-time buyers. By targeting the 1,711 customers in the higher engagement segments with tailored promotions and coupons, the e-commerce platform can potentially increase its conversion rate and revenue.

In summary, the VDAR model successfully segments non-purchasing customers into distinct groups, helping e-commerce businesses identify potential first-time buyers and customers who may need further engagement. This segmentation is particularly valuable for targeting promotional efforts and optimizing marketing strategies based on customer behavior insights.

F. PREDICTION OF CUSTOMER SEGMENTATION USING NEURAL NETWORKS

1) MODEL VALIDATION USING REPEATED STRATIFIED K-FOLD CROSS VALIDATION

To ensure the robustness and reliability of the proposed RFMVDA model, we employed the Repeated Stratified K-Fold Cross Validation method, which is particularly

suitable for classification tasks. This technique involves performing multiple iterations of cross-validation with different shuffles of the data, which helps in assessing the model's performance more comprehensively.

For this validation, the `RepeatedStratifiedKFold` function from the `sklearn` library was used, as it is designed for classification models. This method allowed for a detailed evaluation of how well the model generalizes to unseen data, beyond the initial training set.

As shown in Figure 3, the cross-validation results indicate an average accuracy of 96.9%, which confirms the model's strong generalization capability across different data folds.

2) EXPLORATION OF DIFFERENT DNN ARCHITECTURES AND HYPERPARAMETERS

In order to optimize the model's performance and address potential overfitting risks, we explored various DNN architectures and hyperparameters. The experiments involved tuning different configurations, such as the number of layers, the number of neurons per layer, activation functions, learning rates, batch sizes, and dropout rates.

To systematically evaluate these configurations, we utilized a hyperparameter grid search in combination with repeated stratified K-fold cross-validation. The grid search was configured as follows:

```
param_grid = {
    'layers': [[128], [64, 64], [128, 128]], #
    'activation': ['relu', 'sigmoid'], #
    'optimizer': ['adam', 'sgd'], #
    'dropout_rate': [0.0, 0.2, 0.5], #
    'batch_size': [32, 64], #
    'epochs': [50, 100, 200] #
}

# Repeated Stratified \text{K-Fold} Cross
# Validation setup
rskf = RepeatedStratifiedKFold(n_splits=3,
    n_repeats=1, random_state=42)

# GridSearchCV setup and model training
grid = GridSearchCV(estimator=model, param_grid=
    param_grid, n_jobs=-1, cv=rskf)
grid_result = grid.fit(x_data, y_data_encoded)

# Output the best score and corresponding
# parameters
print('Best Score: {grid_result.best_score_}')
print('Best Params: {grid_result.best_params_}')
```

This approach allowed us to identify the optimal combination of hyperparameters, which resulted in a best score of 0.9725614984830221 with the following configuration: activation function 'relu', batch size 32, dropout rate 0.2, 200 epochs, layers [128, 128], and optimizer 'adam'.

The experimental results provided valuable insights into the impact of various hyperparameters on the model's

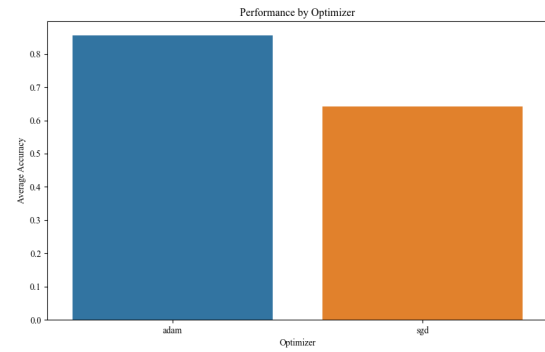


FIGURE 4. Performance by optimizer.

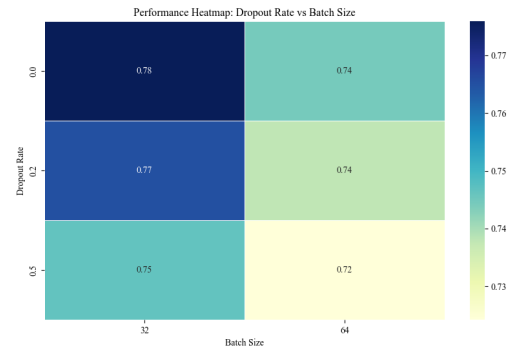


FIGURE 5. Performance Heatmap: Dropout Rate vs Batch Size.

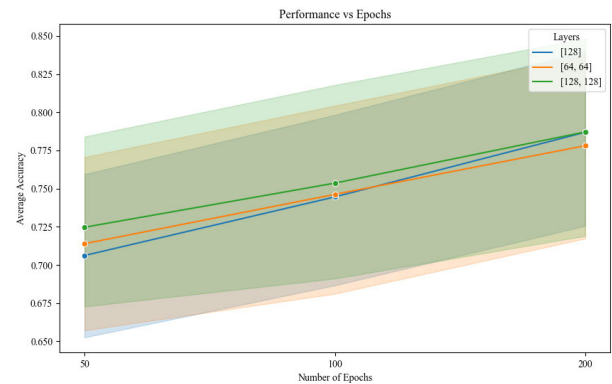


FIGURE 6. Performance vs. Epochs.

performance. The following figures illustrate the performance comparison between different optimizers and dropout rates:

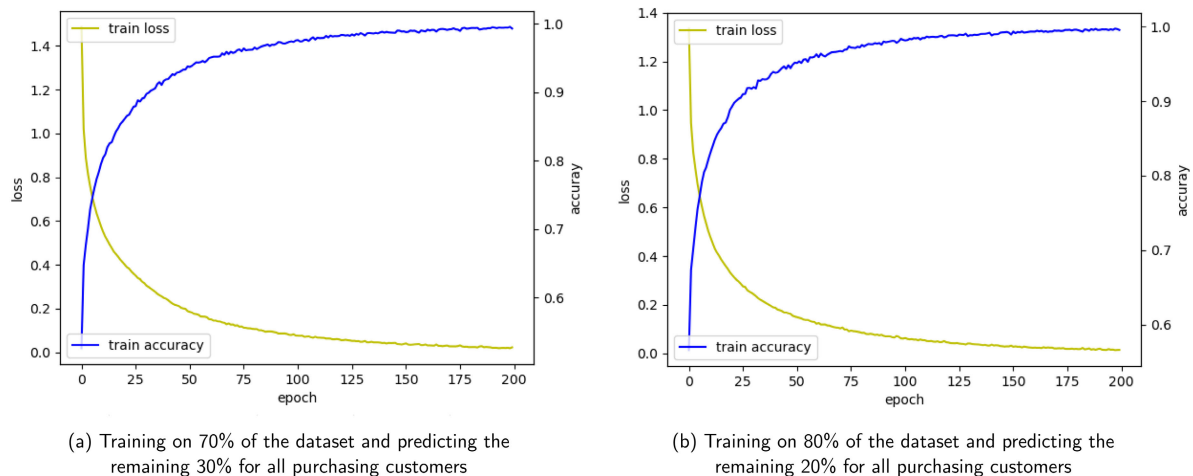
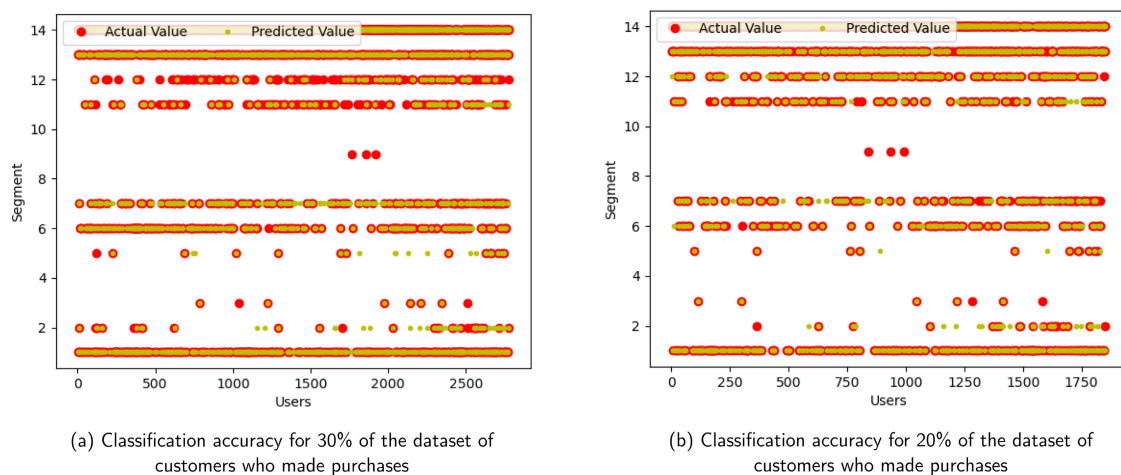
As shown in Figure 4, the Adam optimizer provided better average accuracy compared to SGD, making it the preferable choice for this model.

The heatmap in Figure 5 illustrates the interaction between dropout rates and batch sizes, with a dropout rate of 0.0 and a batch size of 32 achieving the highest accuracy.

Finally, Figure 6 depicts the relationship between the number of epochs and performance, indicating that increasing the epochs generally improves accuracy, particularly for deeper networks with more layers.

TABLE 9. Performance metrics for RFMVDA model.

Metric	Training 70% of the dataset and predicting 30%	Training 80% of the dataset and predicting 20%
Training Accuracy	99.34%	99.54%
Prediction Accuracy	88.26%	92.98%
RMSE	1.6251	1.1778
Macro F1 Score	0.877	0.7923
Micro F1 Score	0.8870	0.9390

**FIGURE 7.** Training and Validation Loss: Training (yellow) and validation (blue) loss over 200 epochs using optimal hyperparameters. The validation loss shows minimal fluctuation, indicating good generalization.**FIGURE 8.** Training and Validation Accuracy: Accuracy trends over 200 epochs for training datasets, showing predicted values closely matching actual values, indicating high performance and minimal overfitting.

3) EVALUATION OF OPTIMAL HYPERPARAMETERS AND MODEL PERFORMANCE

In the previous sections, we explored various DNN architectures and hyperparameters to determine the optimal settings for customer segmentation in the e-commerce environment. After extensive experiments using Repeated Stratified K-Fold Cross Validation, the optimal configuration was identified as follows:

- Optimizer: Adam
- Dropout Rate: 0.2
- Batch Size: 32
- Number of Epochs: 200
- Network Architecture: Two hidden layers, each with 128 neurons, using ReLU activation

Using these optimal hyperparameters, the model was retrained on 80% of the dataset and tested on the remaining

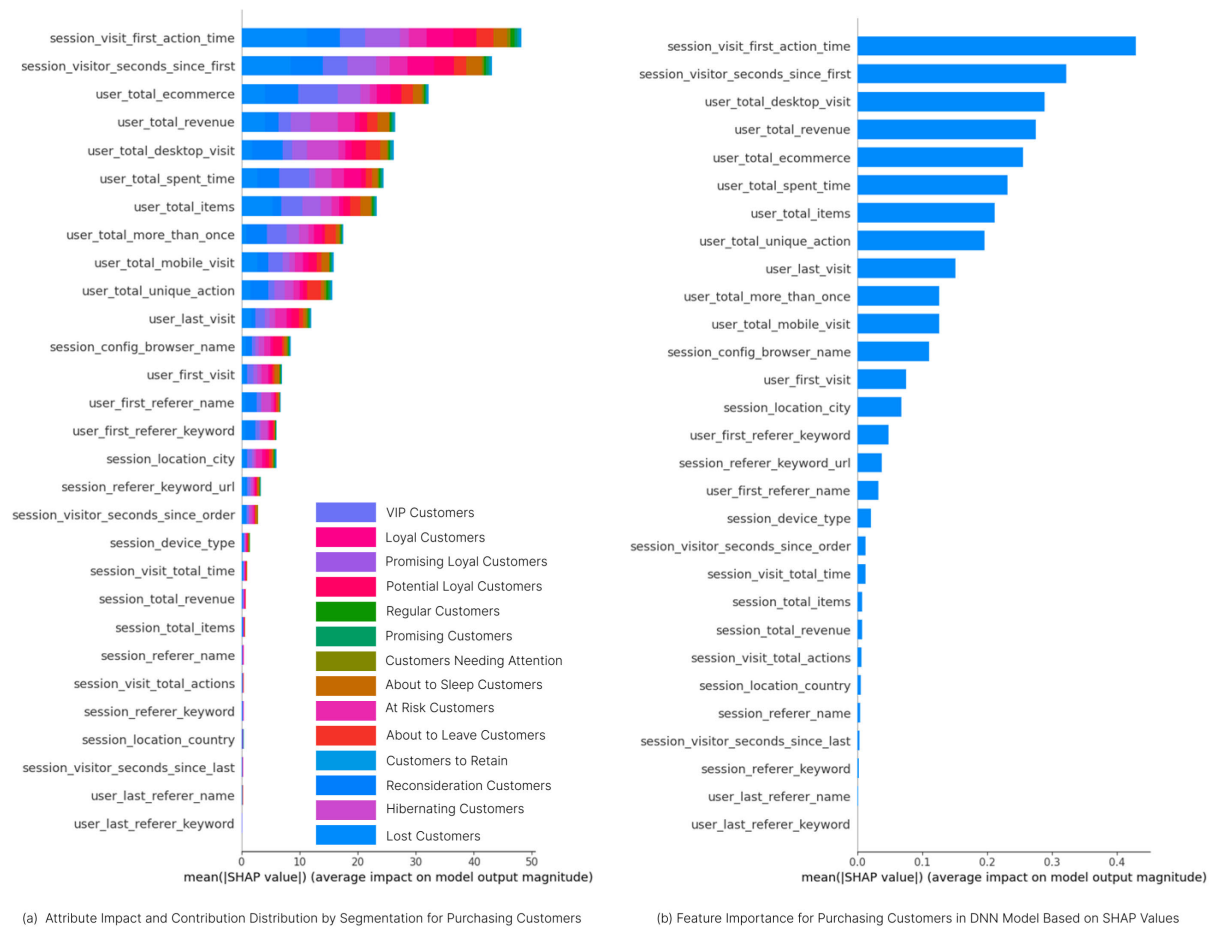


FIGURE 9. SHAP Feature Importance for Customer Segmentation in DNN Model: (a) Segmentation Distribution by Customer Type, (b) Overall Feature Impact on Purchasing Customers.

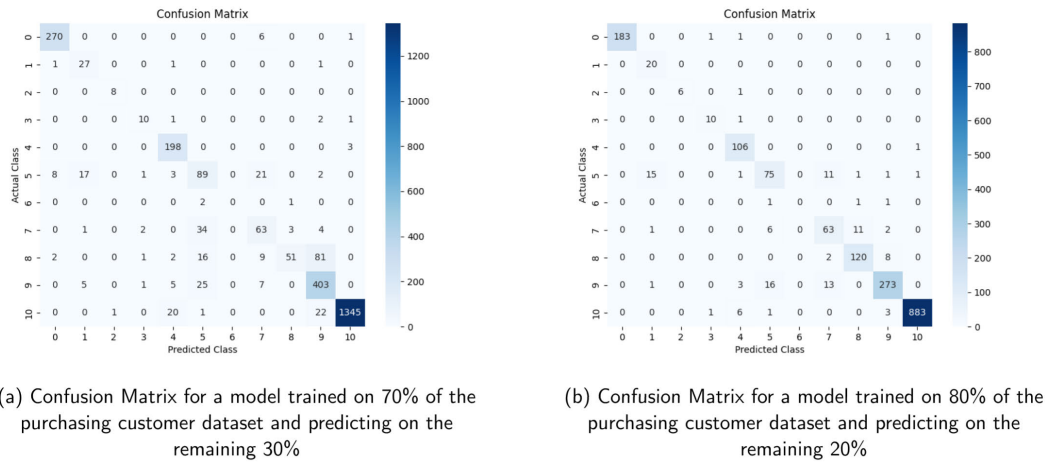


FIGURE 10. Confusion matrix for different training and prediction splits.

20%. As summarized in Table 9, the model achieved a training accuracy of 99.54% and a prediction accuracy of 92.98%. The trend of training and prediction accuracy over epochs is depicted in Figure 7, showing consistent accuracy improvements and demonstrating the model’s robust learning

capacity. Further validation of the model’s performance is provided in Figure 8, which displays loss and accuracy metrics during training and testing. These results confirm minimal overfitting, suggesting that the selected model is well-suited for this classification task. To understand each

feature's contribution to the model's predictions, we analyzed the importance of individual input features using SHAP (SHapley Additive exPlanations) values. Figure 9 displays the average impact of each feature on customer segmentation outcomes, with session duration, first action time, and the number of desktop visits identified as the most influential factors in determining segmentation results.

Additional performance metrics were calculated to offer a more comprehensive evaluation. Specifically, the RMSE (Root Mean Squared Error), Macro F1 Score, and Micro F1 Score were included to provide further insights into the model's accuracy and robustness. The RMSE values of 1.6251 for the 70%-30% split and 1.1778 for the 80%-20% split indicate a low average error magnitude, reflecting high prediction precision. The Macro F1 Score values (0.877 for 70%-30% and 0.7923 for 80%-20%) demonstrate balanced performance across classes, while the Micro F1 Score (0.8870 for 70%-30% and 0.9390 for 80%-20%) underscores the model's effectiveness in accurately classifying individual instances. These metrics are detailed in Table 9.

To further assess the model's robustness and generalization capabilities across different data splits, confusion matrices for the 70%-30% and 80%-20% splits were generated, as illustrated in Figure 10. The confusion matrix for the 70%-30% split in Figure 10(a) illustrates the distribution of correct and incorrect classifications across all classes, identifying areas for potential misclassification. The matrix for the 80%-20% split in Figure 10(b) shows improved accuracy with the larger training dataset, resulting in a more balanced prediction distribution across classes. This comparison provides valuable insights into the model's consistency and robustness under varying data conditions.

In conclusion, the selected DNN architecture and hyperparameters have proven effective for accurate customer segmentation in e-commerce environments. These findings have practical implications, as they can contribute to refining marketing strategies and enhancing personalized customer experiences.

V. CONCLUSION

With the rapid growth of the e-commerce market, traditional customer classification methods, such as the RFM model, face limitations in capturing diverse and continuous customer behaviors. These methods primarily focus on purchasing customers and fail to analyze visitors who interact with platforms but do not make purchases. To address these shortcomings, this paper introduced the RFMVDA (Recency, Frequency, Monetary, Visits, Durations, Actions) model, which extends the RFM framework by incorporating behavioral dimensions such as visit frequency, session duration, and action counts. The RFMVDA model enables a more detailed and comprehensive understanding of purchasing customer interactions in dynamic e-commerce environments.

Recognizing the need to analyze non-purchasing customers as well, this study proposed the VDAR (Visits, Durations, Actions, Referral Keyword) model. Unlike the

RFM-based models that focus solely on transactional data, the VDAR framework provides valuable insights into the engagement levels of visitors who do not complete purchases. By identifying patterns in non-purchasing behaviors, such as visit frequency and referral keywords, the VDAR model enables businesses to uncover opportunities for re-engagement strategies and better understand early indicators of potential future conversions.

The performance of the RFMVDA model was validated using a Deep Neural Network (DNN), achieving a segmentation prediction accuracy of 92.98% and a training accuracy of 99.54%, as shown in Table 9. Additional metrics, including RMSE, Macro F1 Score, and Micro F1 Score, further demonstrated its robustness. SHAP analysis in Figure 9 highlighted the importance of session duration, first action time, and desktop visits as key features. The confusion matrices in Figure 10 confirmed the model's consistency across varying data splits.

Although the results are promising, this study recognizes several limitations. The current analysis relies on historical data collected over a three-month period, which may not fully capture seasonal or long-term behavioral trends. To address this constraint, future research will incorporate data from longer observation windows and multiple e-commerce platforms, allowing the RFMVDA and VDAR models to be evaluated under broader and more diverse conditions. In addition, direct comparisons with other machine learning algorithms, such as Random Forests and Gradient Boosting, were not performed in this study but will be systematically pursued in future work. By conducting benchmark experiments under standardized conditions, these comparative analyses will not only highlight the unique advantages of the RFMVDA framework but also pinpoint areas for further refinement, ultimately enhancing its real-world applicability.

Future research will also investigate how refined segmentation models like RFMVDA can be quantitatively linked to key financial metrics, including customer lifetime value, revenue growth, and retention rates. By examining these relationships in practical e-commerce environments, we aim to clarify how enhanced segmentation strategies can drive long-term profitability and support evidence-based decision-making. This approach will help businesses allocate resources more effectively and prioritize customer-centric programs based on a clear understanding of potential returns.

In conclusion, the RFMVDA and VDAR models, combined with a DNN architecture, represent a significant advancement in customer behavior classification for e-commerce environments. These frameworks address the limitations of traditional approaches while providing actionable insights to enhance customer engagement, improve retention strategies, and support sustainable business growth. By capturing both purchasing and non-purchasing user patterns, they enable businesses to tailor interventions more precisely, ultimately unlocking new opportunities for personalized marketing and data-driven decision-making.

REFERENCES

- [1] *Big Data*, Ministry SMEs Startups of the Republic of Korea, Sejong-si, South Korea, 2022.
- [2] J.-G. Lee and M. Kang, "Geospatial big data: Challenges and opportunities," *Big Data Res.*, vol. 2, no. 2, pp. 74–81, Jun. 2015.
- [3] C. Bull, "Strategic issues in customer relationship management (CRM) implementation," *Bus. Process Manage. J.*, vol. 9, no. 5, pp. 592–602, Oct. 2003.
- [4] Z. Soltani and N. J. Navimipour, "Customer relationship management mechanisms: A systematic review of the state of the art literature and recommendations for future research," *Comput. Hum. Behav.*, vol. 61, pp. 667–688, Aug. 2016.
- [5] S.-Y. Kim, T.-S. Jung, E.-H. Suh, and H.-S. Hwang, "Customer segmentation and strategy development based on customer lifetime value: A case study," *Expert Syst. Appl.*, vol. 31, no. 1, pp. 101–107, Jul. 2006.
- [6] H. Wilson, E. Daniel, and M. McDonald, "Factors for success in customer relationship management (CRM) systems," *J. Marketing Manage.*, vol. 18, nos. 1–2, pp. 193–219, Feb. 2002.
- [7] I. J. Chen and K. Popovich, "Understanding customer relationship management (CRM): People, process and technology," *Bus. Process Manage. J.*, vol. 9, no. 5, pp. 672–688, Oct. 2003.
- [8] M. Kihn and C. B. O'Hara, *Customer Data Platforms: Use People Data To Transform Future Marketing Engagement*. Hoboken, NJ, USA: Wiley, 2020.
- [9] J. T. Wei, S. Y. Lin, and H. H. Wu, "A review of the application of RFM model," *Afr. J. Bus. Manage.*, vol. 4, no. 19, p. 4199, 2010.
- [10] H.-H. Wu, S.-Y. Lin, and C.-W. Liu, "Analyzing patients' values by applying cluster analysis and LRFM model in a pediatric dental clinic in Taiwan," *Scientific World J.*, vol. 2014, pp. 1–7, Jun. 2014.
- [11] R. Mahfuza, R. S. Uddin, Y. Rahman, and Md. A. Hai, "A comprehensive framework for superstore bus. With employing effective clustering techniques," in *Proc. 24th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Dec. 2021, pp. 1–6.
- [12] Y.-L. Chen, M.-H. Kuo, S.-Y. Wu, and K. Tang, "Discovering recency, frequency, and monetary (RFM) sequential patterns from customers' purchasing data," *Electron. Commerce Res. Appl.*, vol. 8, no. 5, pp. 241–251, Oct. 2009.
- [13] A. A. Zoeram and A. K. Mazidi, "New approach for customer clustering by integrating the LRFM model and fuzzy inference system," *Iranian J. Manage. Stud.*, vol. 11, no. 2, pp. 351–378, Apr. 2018.
- [14] A. J. Christy, A. Umamakeswari, L. Priyatharsini, and A. Neyaa, "RFM ranking—An effective approach to customer segmentation," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 33, no. 10, pp. 1251–1257, Dec. 2021.
- [15] X. He and C. Li, "The research and application of customer segmentation on e-commerce websites," in *Proc. 6th Int. Conf. Digit. Home (ICDH)*, Dec. 2016, pp. 203–208.
- [16] C. O. Sakar, S. O. Polat, M. Katircioglu, and Y. Kastro, "Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks," *Neural Comput. Appl.*, vol. 31, no. 10, pp. 6893–6908, Oct. 2019.
- [17] Y.-S. Shih and M.-H. Lin, "A lstm approach for sales forecasting of goods with short-term demands in e-commerce," in *Proc. Asian Conf. Intell. Inf. Database Syst.*, 2019, pp. 244–256.
- [18] Y. Issaoui, A. Khat, A. Bahnasse, and H. Ouajji, "An advanced LSTM model for optimal scheduling in smart logistic environment: E-commerce case," *IEEE Access*, vol. 9, pp. 126337–126356, 2021.
- [19] C. Wang, "Efficient customer segmentation in digital marketing using deep learning with swarm intelligence approach," *Inf. Process. Manage.*, vol. 59, no. 6, Nov. 2022, Art. no. 103085.
- [20] M. S. Kasem, M. Hamada, and I. Taj-Eddin, "Customer profiling, segmentation, and sales prediction using AI in direct marketing," *Neural Comput. Appl.*, vol. 36, no. 9, pp. 4995–5005, Mar. 2024.
- [21] K. Sriprasad, S. Palit, B. Pravallika, Manjunatha, R. Lenka, and A. Singla, "Client segmentation and customization in e-commerce: Applications of machine learning from a management perspective," in *Proc. Int. Conf. Commun., Comput. Sci. Eng. (IC3SE)*, May 2024, pp. 1423–1427.
- [22] M. Rajyalaxmi, C. Vijai, K. Srivastava, N. Kalyan, B. Pravallika, and A. Dutt, "Application of machine learning algorithms for customer segmentation in e-commerce management," in *Proc. Int. Conf. Sci. Technol. Eng. Manage. (ICSTEM)*, Apr. 2024, pp. 1–5.



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